



BLM4021 Gömülü Sistemler

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Yıldız Teknik Üniversitesi – Bilgisayar Mühendisliği

Lecture 13 – Internet of Things

- What is IoT?
- Basic Characteristics and Architectures,
- Design Criteria,
- Communication,
- Industry 4.0,
- Some Applications.

Kaynaklar:

- M. A. Iqbal, S. Hussain, H. Xing, M. A. Imran, Enabling The Internet of Things: Fundamentals, Design and Applications, Wiley.
- I. Culic et al, Commercial and Industrial IoT Apps with the Raspberry Pi, Apress.
- L. Bononi, Marco di Felice, Introduction to IoT enabling DR: Technologies, apps, perspectives, Lecture Note.
- A. Bahga, V. Madisetti, Internet Of Things: A Hands-On Approach, Lecture Notes.
- Ala. Al-Fuqaha, Internet of Things: Applications & Prospects.
- M. Cardona, V. K. Solanki, C. E. G. Cena, Internet of Medical Things: Paradigm of Wearable Devices, CRC Press.

Haftalık Konular



Hafta	Teorik	Laboratuvar
1	Giriş ve Uygulamalar, Mikroişlemci, Mikrodenetleyici ve Gömülü sistem kavramlarının açıklanması	Grupların oluşturulması & Kitlerin Testi
2	Bir Tasarım Örneği, Mikroişlemci, Mikrodenetleyici, DSP, FPGA, ASIC kavramları	Kitlerin gruplara dağıtımı + Raspberry Pi Kurulumu
3	16, 32 ve 64 bitlik mikrodenetleyiciler, pipeline	Raspberry Pi ile Temel Konfigürasyon
4	PIC ve MSP430 özellikleri	---- Resmi Tatil ---
5	ARM tabanlı mikrodenetleyiciler ve özellikleri	Uygulama 1 – Raspberry Pi ile Buzzer Uygulaması
6	ARM Komut setleri ve Assembly Kodları-1	Uygulama 2 – Raspberry Pi ile Ivme ve Gyro Uygulaması
7	ARM Komut setleri ve Assembly Kodları-2, Raspberry Pi vers. ve GPIO'ları	Uygulama 3 – Raspberry Pi ile Motor Kontrol Uygulaması
8	Vize Sınavı	
9	Haberleşme Protokolleri (SPI, I2C ve CAN)	Proje Soru-Cevap Saati - 1
10	Sensörlerden Veri Toplama, Algılayıcı, ADC ve DAC	Proje Soru-Cevap Saati - 2
11	Zamanlayıcı, PWM ve Motor Sürme	Proje kontrolü - 1
12	Gerçek Zaman Sistemler ve temel kavramlar	Proje Kontrolü - 2
13	Gerçek zaman İşletim Sistemleri	
14	Nesnelerin İnterneti	
15	Final Sınavı	

For more details -> Bologna page: <http://www.bologna.yildiz.edu.tr/index.php?r=course/view&id=9463&aid=3>

Internet of Things (IoT)

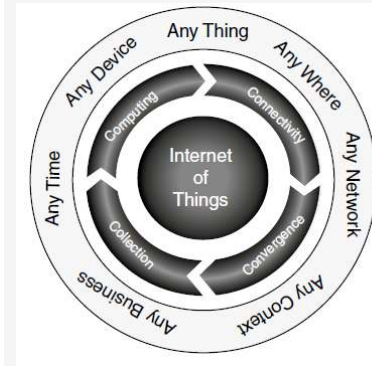
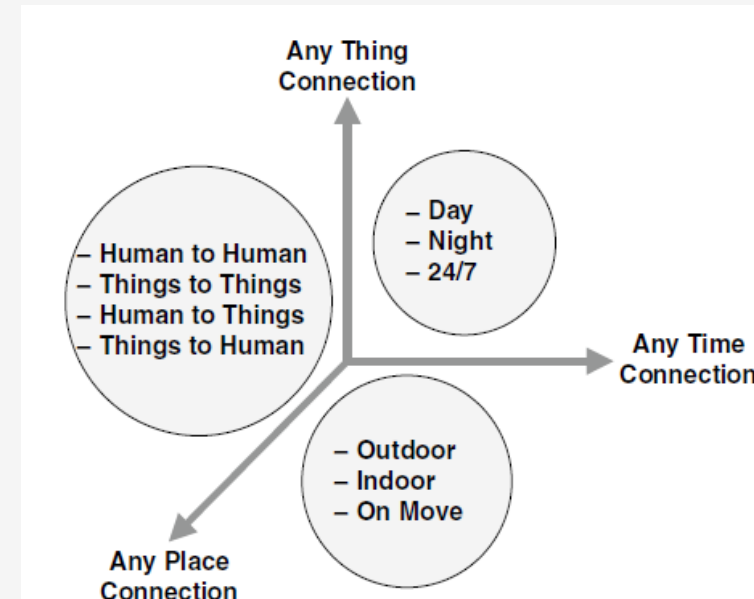


The term Internet of Things (IoT) was first used in 1999 by the British technology pioneer Kevin Ashton, who worked at Procter & Gamble (P&G).

However, it was not until 2009 that this paradigm took off, turning into the third revolutionary wave in computing technology after the personal computer and Internet.

Many Definitions:

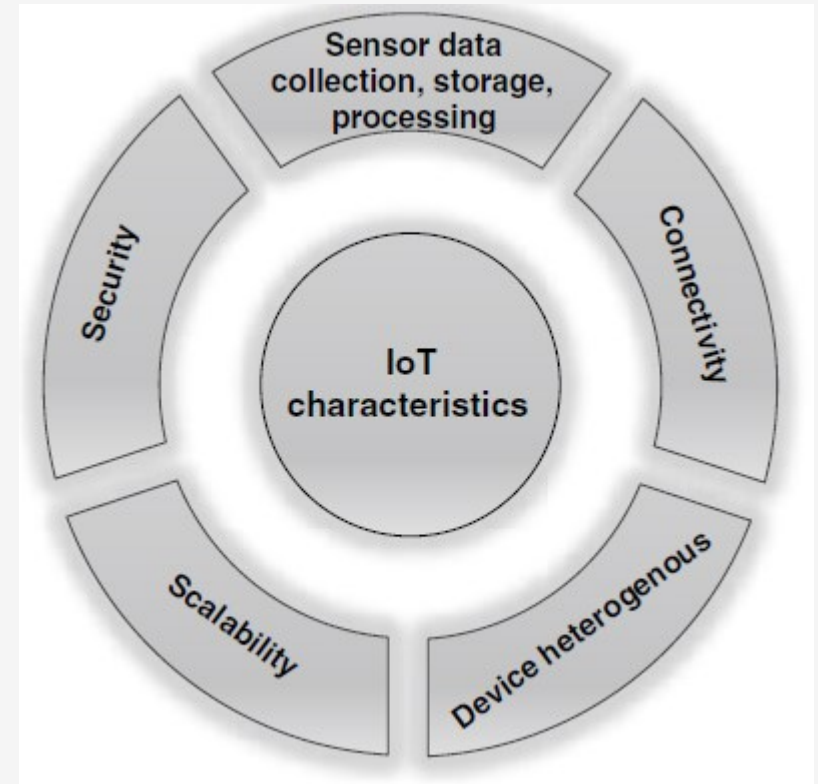
- IoT aims for a **world** where everything, including houses, cars, urban furniture and industrial machinery are connected to the Internet at any given time and place, creating real-time data that facilitates decision making, offering the user comfort while saving time and money. <Book>
- “IoT is type of network that is available anywhere, anytime, by anything and anyone.” -> ITU



IoT Basic Characteristics



IoT characteristic	Description
Sensor Data Acquisition, Storage, Filtering and Analysis	The plethora of distributed Sensors (or smart things) gather observation of physical environment/entity and direct to Cloud for storage and analytics with an ultimate objective to improve business workflow
Connectivity	IoT has made possible the interconnectivity of Physical and Virtual things with the help of the Internet and global communication infrastructure (that is built using wired and wireless technologies)
Device Heterogeneity and Intelligence	The interoperability of devices (based on different hardware and network platforms) with the provisioning of ambient intelligence at the hardware/software level supports intelligent interactions
Scalability	The plethora of IoT devices connectivity shifts human interactions to device interactions
Security	The security paradigm is required to be implemented at the network level as well as the end-devices level to ensure the security of data



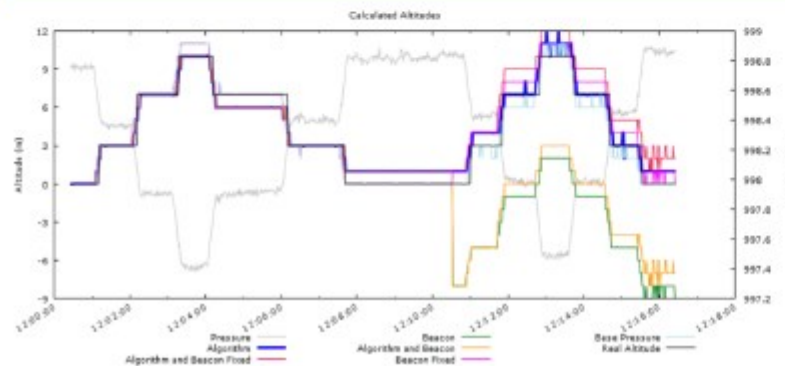
An Example... Mobile Phones

✓ Pervasiveness of Sensing devices (digital data)

- E.g. current smartphones are equipped with a number of embedded sensors

A Mobile Smartphone can provide a lot of context information about you and your activities:

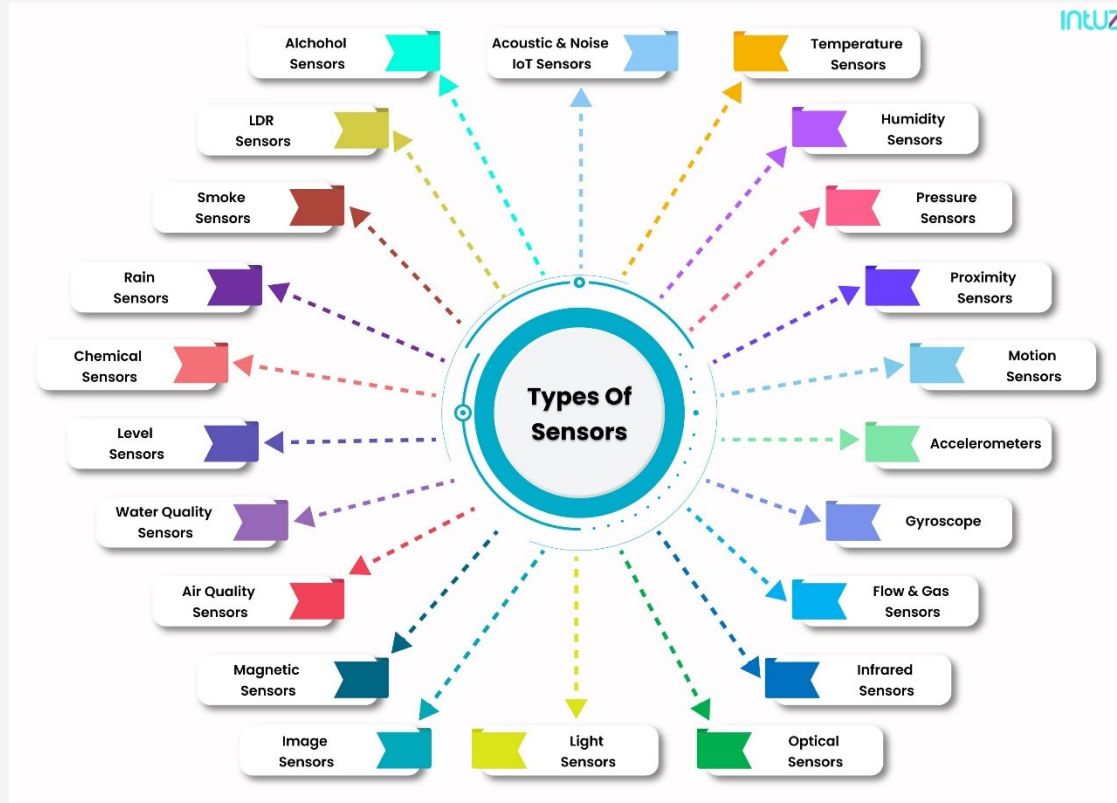
- your mobility (even without GPS)
- your movement between floors in a building
 - your speech
 - your contacts
- your interaction with other objects...
 - YOUR LIFE!



Sensors and Actuators



Sensors

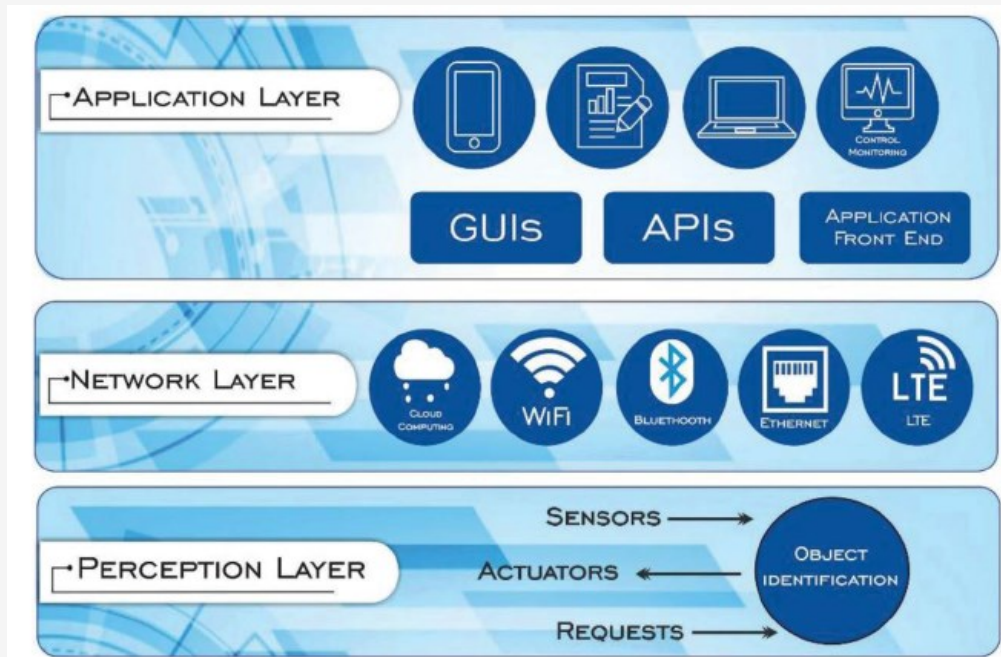


Actuators

Category	Role in the IoT Environment
Linear actuators	Used to enable object motion in a straight line
Motors	Used to rotate mechanisms or objects with high positioning accuracy
Relays	Electromagnet actuators used to power devices
Solenoids	Widely used in home appliances (locking/triggering mechanisms and monitoring systems)

IoT Architectures

In the literature, there are a number of proposed architectures which are typically divided into levels.



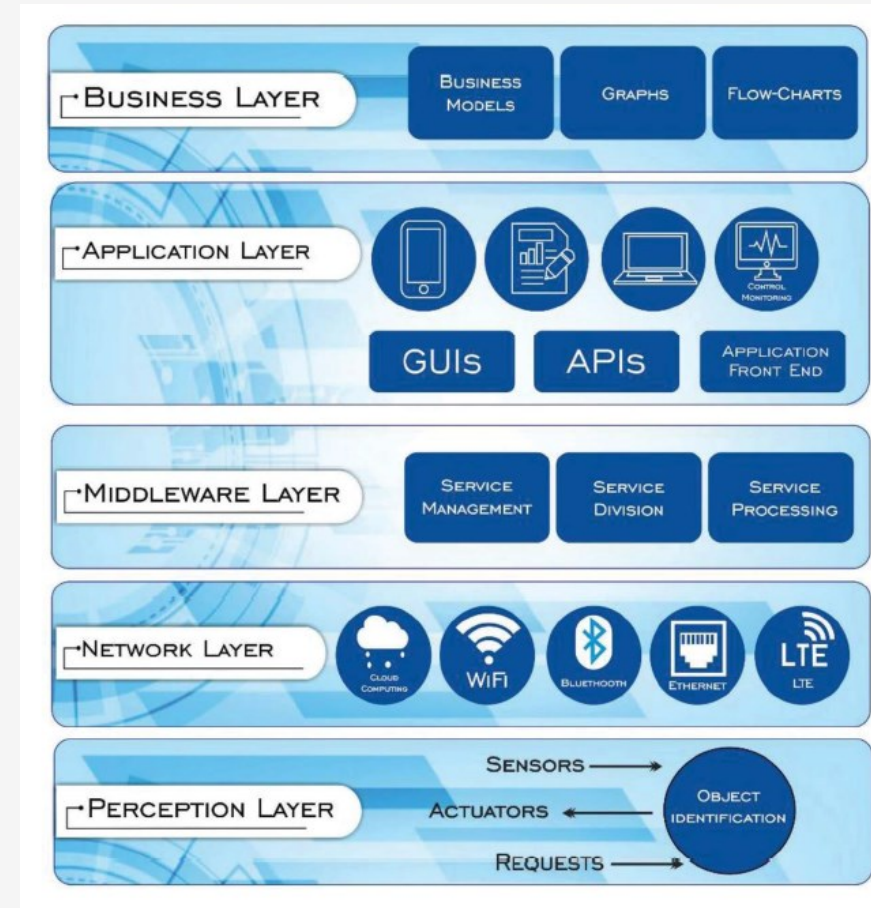
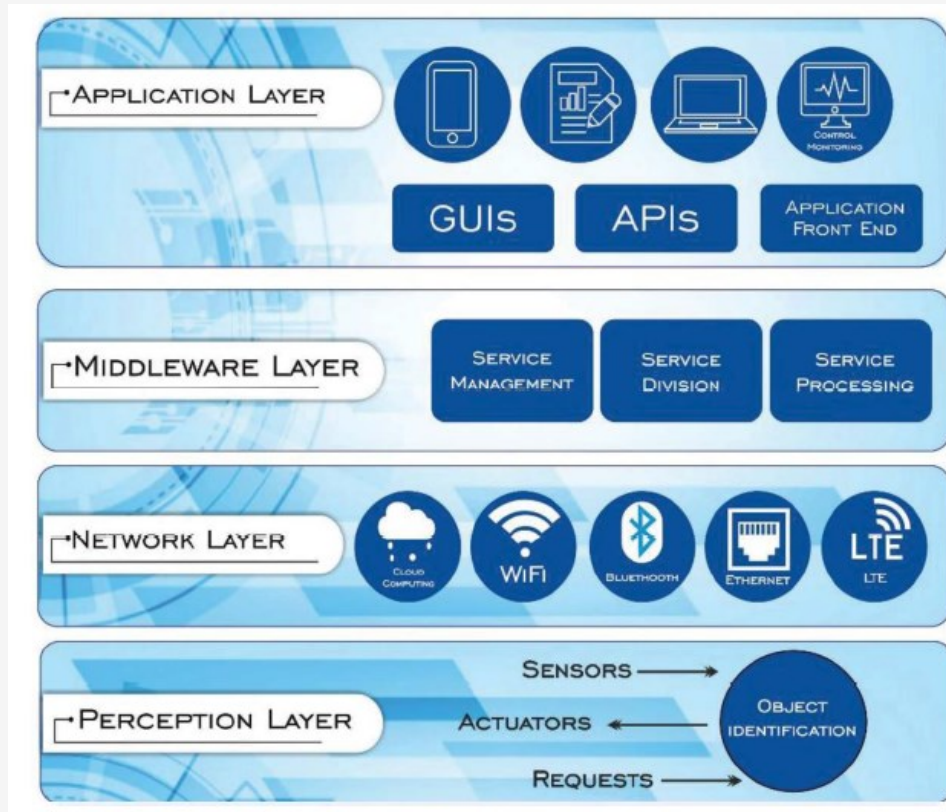
The perception layer: This layer is also known as the terminal layer or the device layer. The perception layer is classified in two sections: the perception node (sensors, actuators, etc.) and the perception network that is responsible for interconnecting the network layer

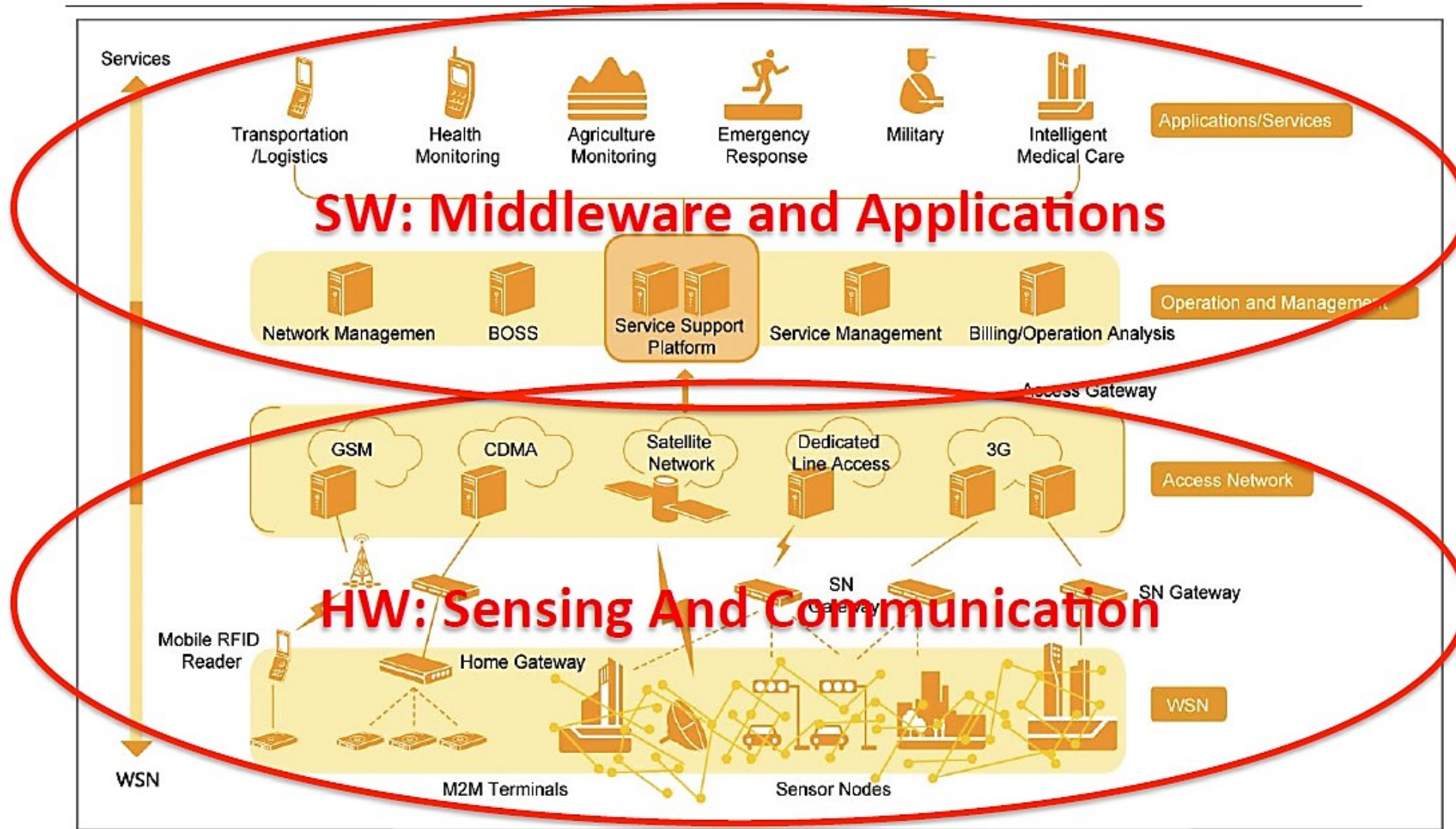
The network layer: Also known as the transport layer. The main function of this layer is transmitting and processing the data coming from the perception layer; in other words, it supports the connectivity of the devices.

The application layer: At this level, various types of developments and user services can be found, which make use of the data collected at the perception level.

IoT Architectures

In the literature, there are a number of proposed architectures which are typically divided into levels.



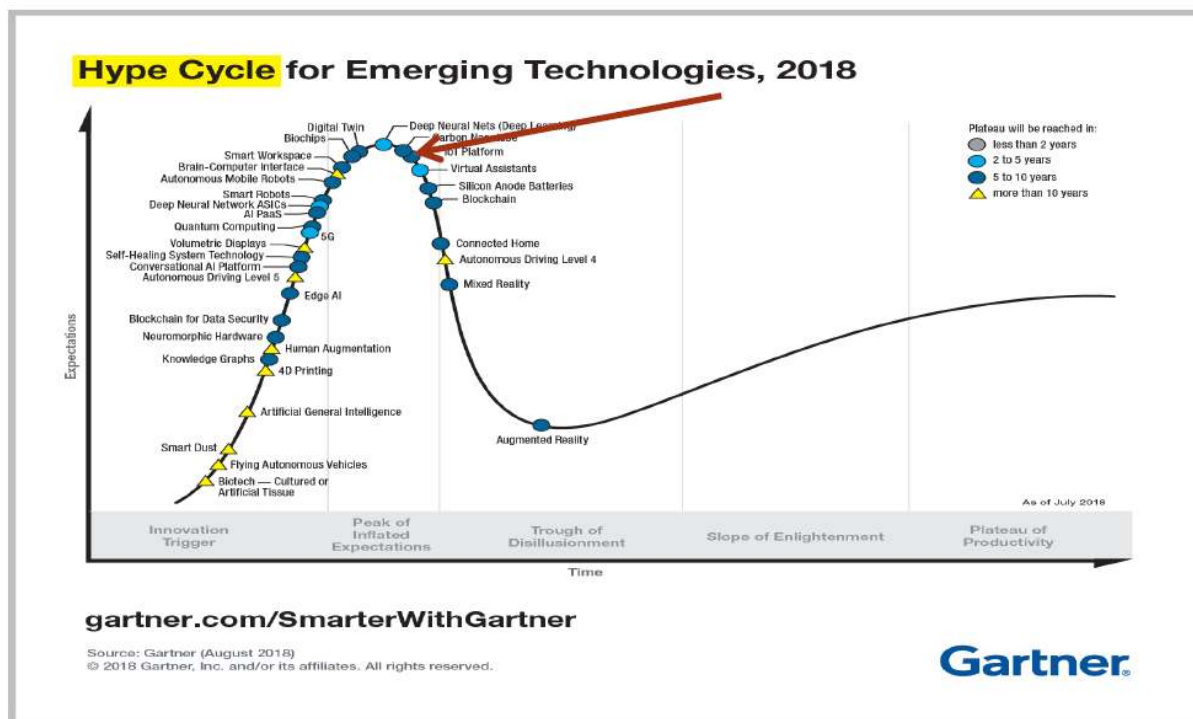


Embedded Systems vs. IoT

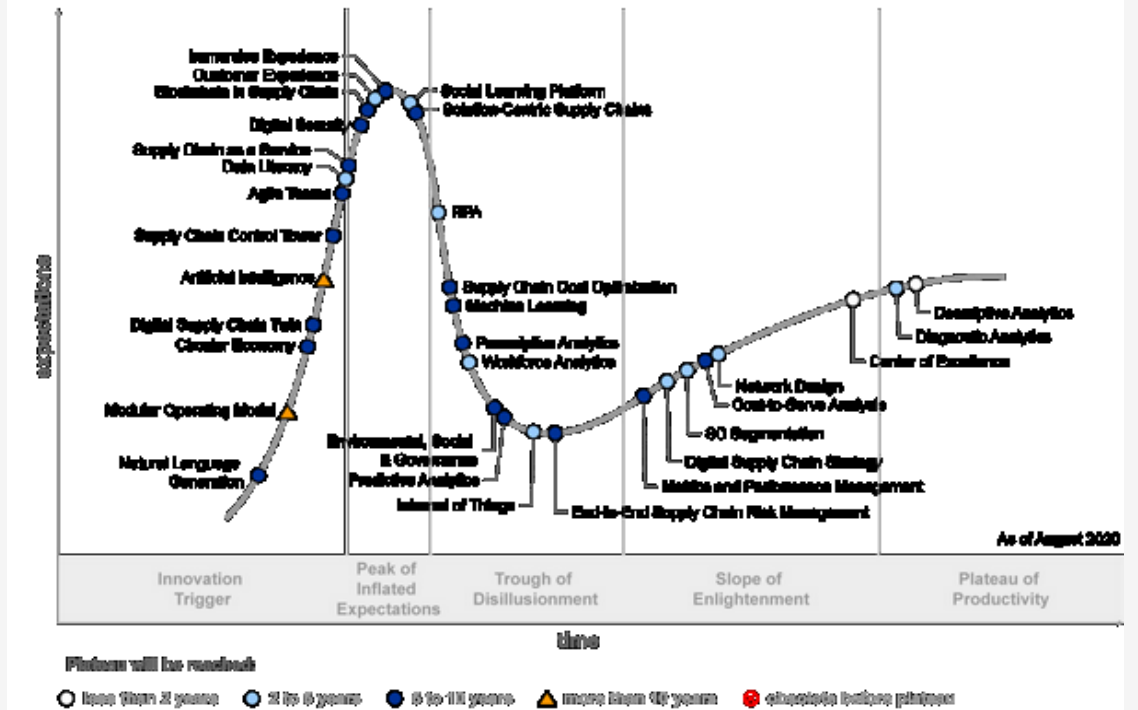


Embedded system	IoT
Embedded systems include electronic devices that are usually standalone in nature and independently run on the Internet	IoT is a system that includes Internet connectivity-reliant devices for communication
Embedded systems are a combination of hardware and software (firmware)	IoT systems are a combination of computer hardware, software, and networking capabilities that are embedded into things of our daily lives
Embedded systems firmware mostly needs no modifications once the device is delivered to the clients	IoT requires continuous update
Example: ECG machine in a healthcare service that analyzes health parameters associated with humans is an example of embedded systems	Example: ECG machine connected to the Internet and able to transfer human health parameters on a remote server is an example of IoT devices
Embedded systems are a subset of IoT	IoT is a broader term including different technologies, i.e. embedded systems, networking, and information technology

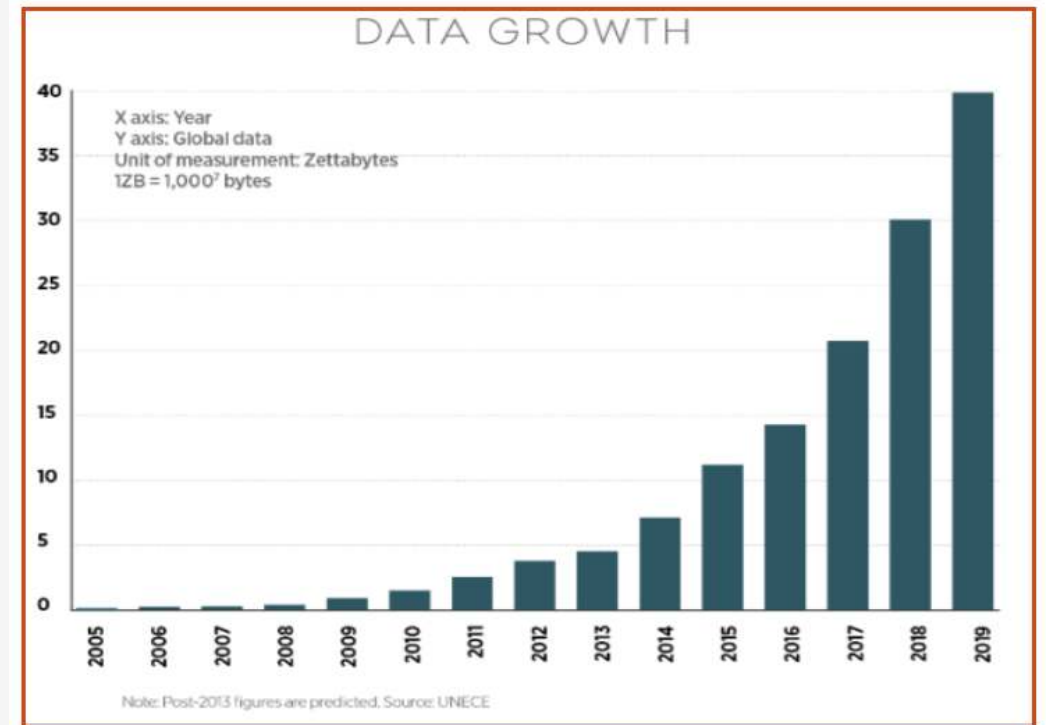
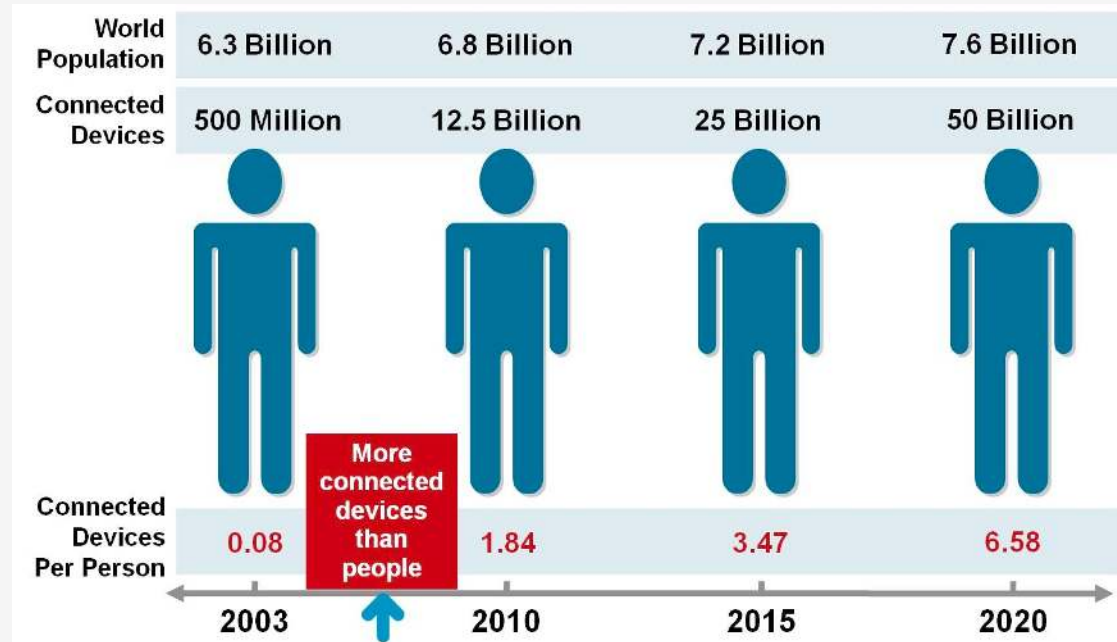
IoT is ... trendy



Hype Cycle for Supply Chain Strategy, 2020



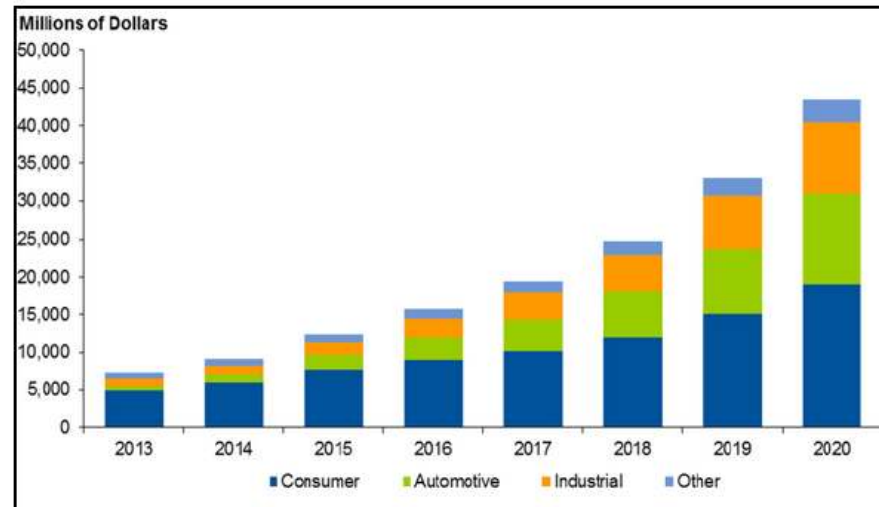
Usage and Data Amount of IoT



Source: <https://www.sdintl.com/2018/04/30/big-data-friend-or-foe/>

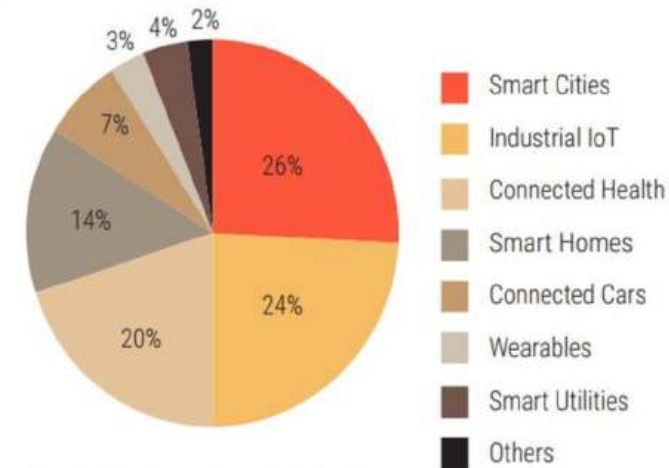


IoT is ... profitable (\$\$\$)



IoT Revenues worldwide, source: Gartner

Global IoT Market Share by Sub-Sector



[Source: GrowthEnabler Analysis]

<https://www.forbes.com/sites/louiscolombus/2017/12/10/2017-roundup-of-internet-of-things-forecasts/#7136007a1480>

IoT Value Chain



Note, the above is not an exhaustive list of companies and any company may have play in more than one component of value chain
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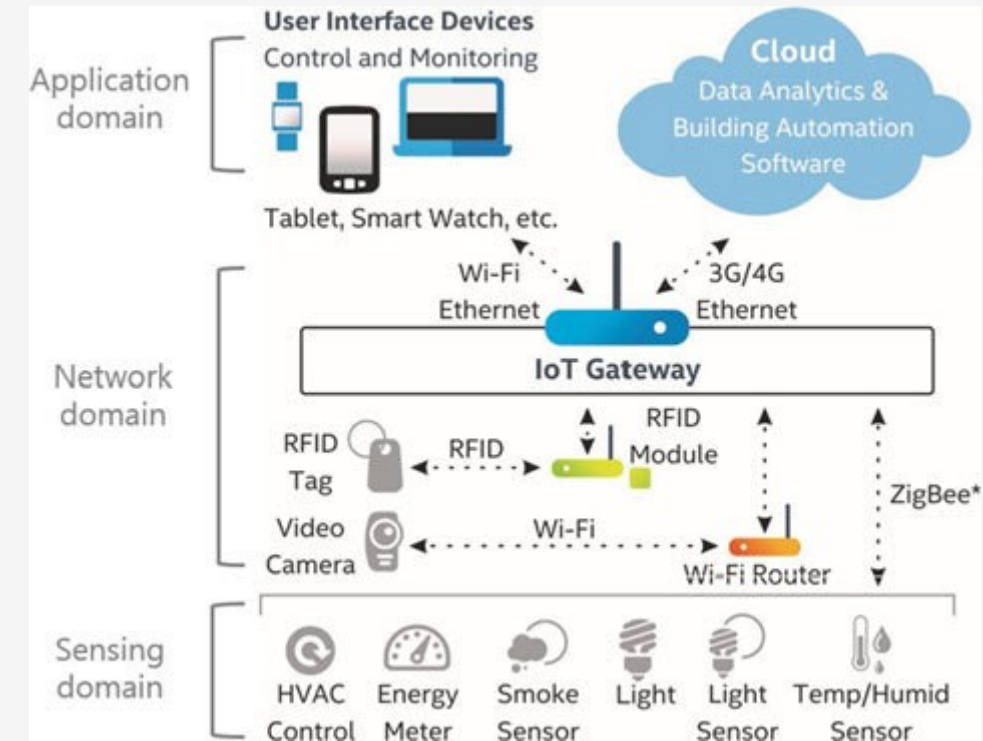
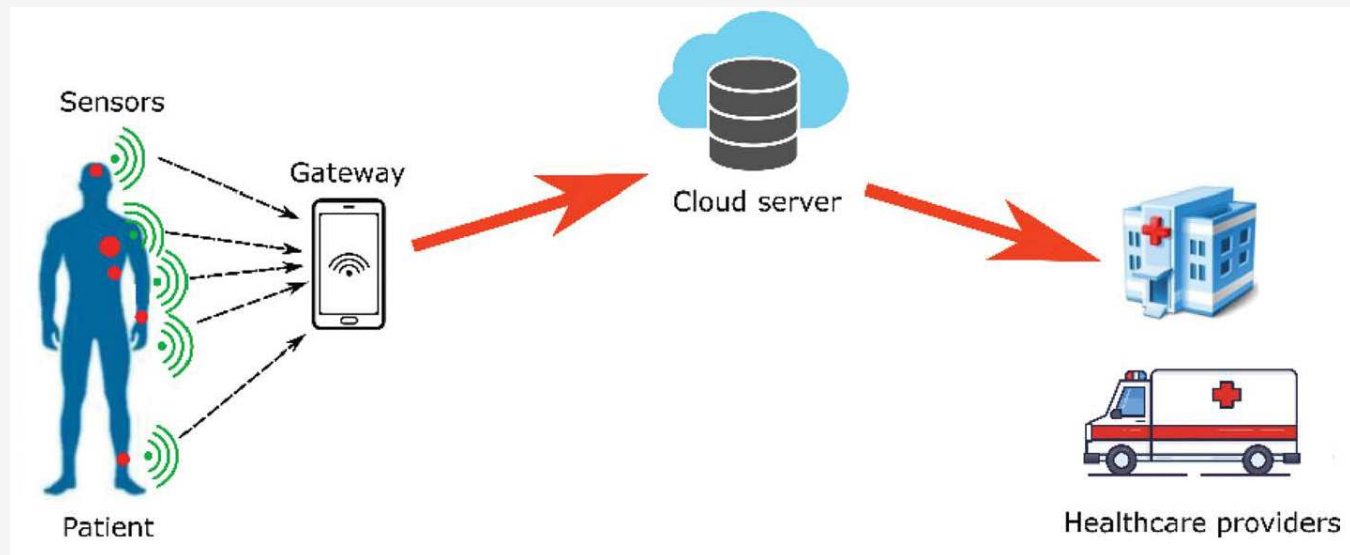
When designing a technology for IoT...



The key considerations are:

- **Data Rate:** How much data needs to be transmitted and how often?
- **Power Availability:** The camera requires a continuous power feed, both for its own operation but also to support the network connection, whereas a smart parking sensor that has been buried in the road surface would have to run off a battery for several years
- **Range:** Use cases vary from short distance, such as homes, offices, cars, and factories, through to long ranges for wider area city networks, deep in-building penetration, or rural use cases such as agriculture.
- **Cost:** Low-cost modules and networks are required for several use cases in order to support the business models to make deployment and operation of devices and sensors commercially viable, so becomes a key factor in real-world deployments.

Ex. IoT based Healthcare Systems



Why do we use gateways?

- WSNs have different protocols and they cannot understand each other. Gateways connecting WSNs to internet

Issues for IoT Nodes



- Limited processing power
- Battery operated, so must be power miser, sleeping capabilities highly desirable
- Robust, deployable in harsh environment
- Weatherproof
- Easy to configure
- Inexpensive, deployable in great numbers

	Local Area Network Short Range Communication	Low Power Wide Area (LPWAN) Internet of Things	Cellular Network Traditional M2M
	40%	45%	15%
😊	Well established standards In building	Low power consumption Low cost Positioning	Existing coverage High data rate
😞	Battery Life Provisioning Network cost & dependencies	High data rate Emerging standards	Autonomy Total cost of ownership
	Bluetooth 4.0  	LoRa	GSM 3G+ / H+ 4G

802.11ah (Sub 1 GHz) WiFi HaLow at 900 MHz

WiFi consumes too much power, so an IoT customised version was developed

Low power, long range Wi-Fi

Up to 1 km range, lower power consumption thanks to a sleep mode

1, 2, 4, 8 and 16 MHz channels

Competes with Bluetooth, 100 kb/s to 40 Mb/s

Bluetooth

Based on IEEE 802.15.1

Smart Mesh

79 channels 1 MHz wide and frequency hopping to combat interference at 2.4 GHz

Used mainly for speakers, health monitors and other short range applications

Bluetooth Low Energy (BLE) or Smart Bluetooth

Based on IEEE 802.15.1

Subset of Bluetooth 4.0, but stemming from an independent Nokia solution

Smart Mesh

Support for IOS, Android, Windows and GNU/Linux

40 channels 2 MHz wide and frequency hopping to combat interference

Used in smartphones, tablets, smart watches, health and fitness monitoring dev.

Zigbee

Based on IEEE 802.15.4, provides the higher layers up to application

Latest standard Zigbee 3.0 issued Dec 2015

Mesh topology

Short range, 20 to 250 kbps

2.4 GHz, 915 MHz or 868 MHz

Channels 2 MHz with Direct Sequence Spread Spectrum media access

Cheaper than Bluetooth

Popularity of Low Power Wide Area Network



Optimized for IoT and Machine to Machine (M2) applications

Trade throughput for coverage (up to several kilometers)

Star topology



9 miles



50 kbps



10 years

Long Range

Low Power

Low Data Rate

Existing Technologies for Communication



- ZigBee / IEEE 802.15.4,
- Bluetooth,
- RF-ID,
- Wi-Fi,
- WiMax,
- 4G (Cellular),
- 5G (Cellular),
- LoRA (LPWAN),
- SigFox (LPWAN)

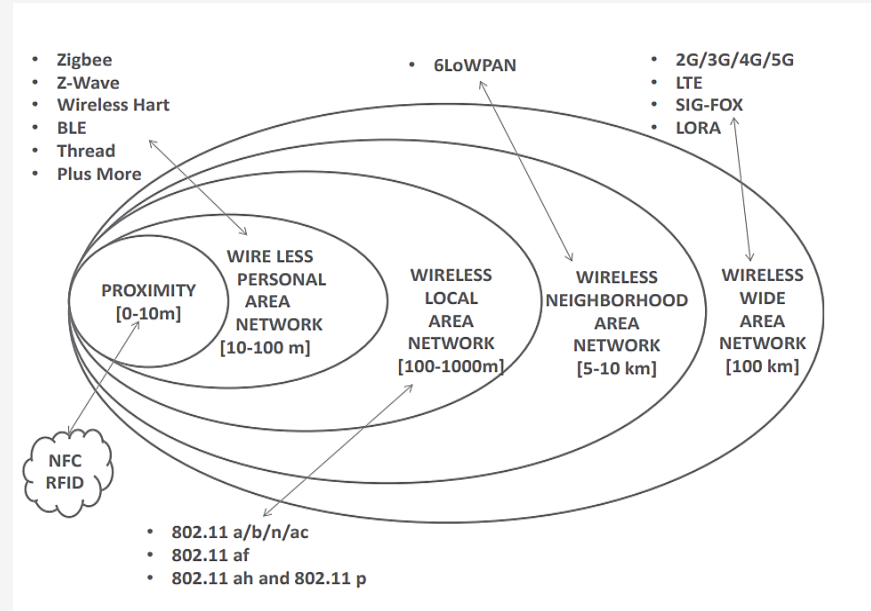


Table 1.1: IoT Communication technologies.

Range	Technology	Standard	Frequency	Bit rate	Coverage
short	WiFi	IEEE 802.11	2.4/5 GHz	500Mbps	50 m
	Zigbee	IEEE 802.15.4	24 GHz	250kbps	100m
	Z-Wave	ZAD12837	900 MHz	40 kbps	50 m
	RFID	ISO/IEC 18000	ISM Band	40 kbps	<2 m
	NFC	ISO/IEC 18092	13.56 MHz	424 kbps	<20 cm
	Bluetooth LE	IEEE 802.1	2.4 GHz	1 Mbps	50 m
	Thread	IEEE 802.15.4	2.4 GHz	250 kbps	<100 m
	Home Plug	IEEE1901	<100 MHz	10-500 Mbps	<100 m
	DSRC	IEEE 802.15P	5 GHz	4Mbps	300 m
Long	Neul SigFox	Neul Sigfox	458 MHz	10 kbps	10 km
	6LoWPAN	RFC6282	900 MHz	1 kbps	10 km
	GSM/ 3G/ 4G/ 5G	GSM, UMTS, HSPA, LTE	ISM Band 900/ 1800/ 1900/ 2100 MHz	N/A	N/A
	WiMax	IEEE 802.16	2.3, 2.5 and 3.5 GHz	10 Mbps	50 km
				410 Mbps	510 km

IoT is ... chaotic



Table 2
Capillary IoT technologies. Some of the data were cross-checked with [87].

Name	Spectrum	Bandwidth	Peak DR	Range	Topology	PHY Modulation	MAC Access
BLE	2.4 GHz	2 MHz	1 Mbps	100 m	Star	GFSK (FHSS)	TDMA
Thread 6LoWPAN	2.4 GHz	5 MHz	250 kbps	10–75 m	Mesh	OQPSK (DSSS)	CSMA/CA
ZigBee	2.4 GHz	2 MHz	250 kbps	10–75 m	All	OQPSK (DSSS)	S-CSMA/CA
ZigBee	915 MHz	1.2 MHz	40 kbps	10–75 m	All	BPSK (DSSS)	S-CSMA/CA
ZigBee	868 MHz	600 kHz	20 kbps	10–75 m	All	BPSK (DSSS)	S-CSMA/CA
WirelessHART	2.4 GHz	3 MHz	250 kbps	30–90 m	Mesh	OQPSK (DSSS)	TDMA
ISA 100.11a	2.4 GHz	5 MHz	250 kbps	30–90 m	Mesh	OQPSK (DSSS)	TDMA
Z-Wave	868/908 MHz	200 kHz	9.6–40 kbps	30–100 m	Mesh	FSK	TDMA
Z-Wave 400	2.4 GHz	–	200 kbps	30–100 m	Mesh	FSK	TDMA
INSTEON	908 MHz	–	38.4 kbps	45 m	Mesh	FSK	TDMA
EnOcean	868/315 MHz	62.5 kHz	125 kbps	30 m	Mesh	ASK, FSK	TDMA
D7AP Hi-Rate	433/868/915 MHz	200 KHz	166.67 kbps	10 m	Tree	GFSK	CSMA/CA
D7AP	433/868/915 MHz	200 KHz	55.55 kbps	100 m	Tree	GFSK	CSMA/CA
DECT ULE	1.8/1.9 GHz	1.728 MHz	1152 kbps	70–300 m	Star	GFSK	TDMA

Table 3
LPWAN technologies operating in unlicensed bands. Some of the data were cross-checked with [87].

Name	Spectrum	Bandwidth	Peak DR UL	Peak DR DL	Range	PHY Modulation	MAC Access
D7AP Lo-Rate	433/868/915 MHz	25 kHz	9.6 kbps	9.6 kbps	~5 km	GFSK	CSMA/CA
SigFox	868–915 MHz	192 kHz	~100 bps	~100 bps	>20 km	GFSK/DBPSK (UNB)	ALOHA
Ingenu MN	2.4 GHz	1 MHz	~30 kbps	~30 kbps	~15 km	FSK, PSK (DSSS)	RPMA
LoRa	868–915 MHz	125 kHz	~50 kbps	~50 kbps	~11 km	CSS	ALOHA
Weightless-N	868 MHz	200 Hz (?)	~100 kbps	–	~5 km	DBPSK (UNB)	S-ALOHA
Weightless-P	868 MHz	12.5 kHz	~100 kbps	100 kbps	~2 km	GMSK, OQPSK (UNB)	FDMA, TDMA
Weightless-W	470–790 MHz	6–8 MHz	~10 Mbps	~10 Mbps	~10 km	DBPSK/QPSK /16-QAM (DSSS)	FDMA, TDMA

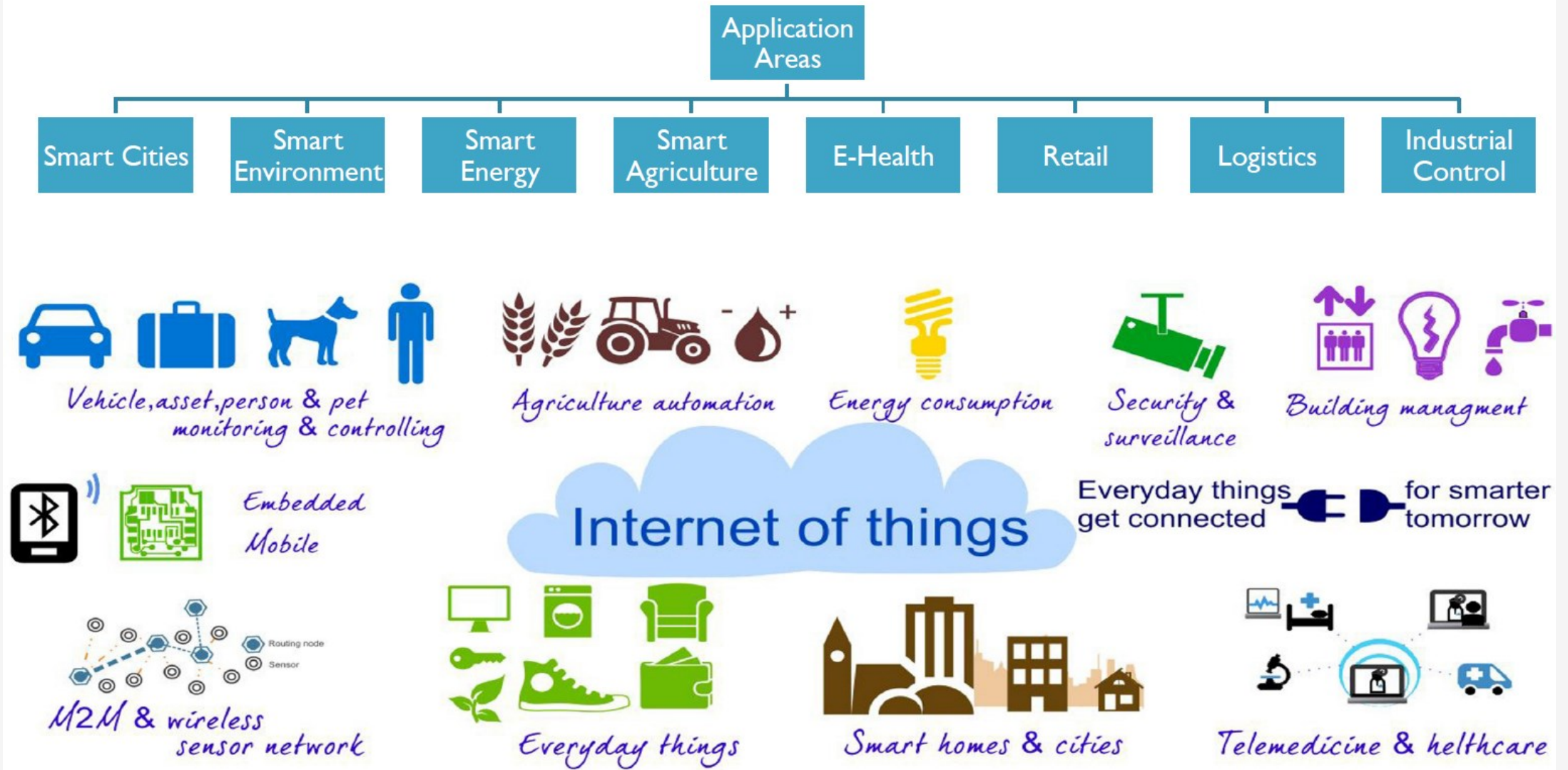
Table 4
Cellular IoT technologies operating in licensed bands. Some of the data were cross-checked with [87].

Name	Spectrum	Bandwidth	Peak DR UL	Peak DR DL	Range	Modulation	Access
EC-GSM	700–900 MHz	200 kHz	~10 kbps	~10 kbps	~15 km	GMSK	TDMA
LTE-M	700–900 MHz	1.4 MHz	~1 Mbps	~1 Mbps	~11 km	QPSK, 16-QAM, 64-QAM	OFDMA
NB-LTE-M	700–900 MHz	200 kHz	~144 kbps	~200 kbps	~15 km	QPSK, 16-QAM, 64-QAM	OFDMA
NB-CIoT	800–900 MHz	180 kHz	~36 kbps	~45 kbps	~15 km	BPSK, QPSK, 16-QAM	OFDMA

Main wireless communication technologies on the market, used to enable Machine-to-Machine communication on IoT scenarios

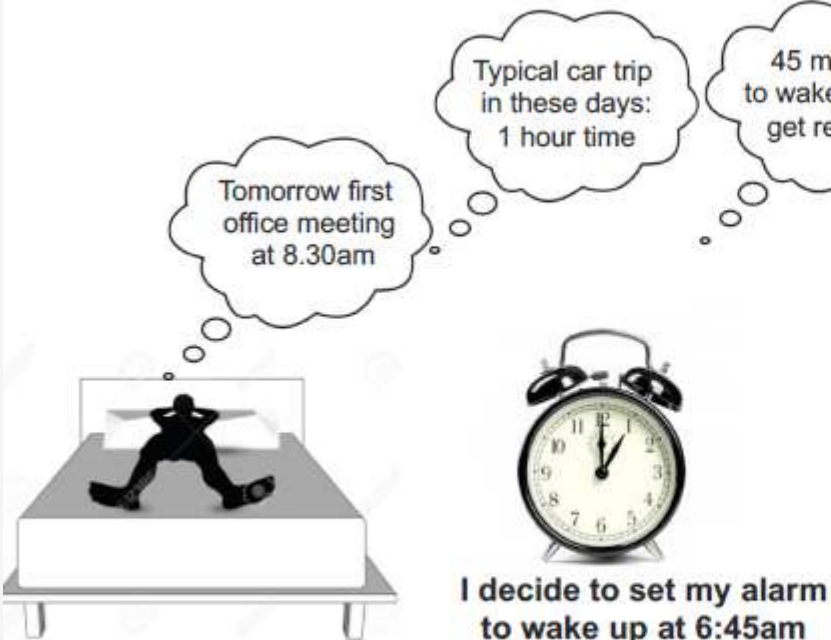
F. Montori, L. Bedogni, M. Di Felice, L. Bononi, Machine-to-Machine Wireless Communication Technologies for the Internet of Things: Taxonomy, Comparison and Open Issues, Pervasive and Mobile Computing (Elsevier), 2018.

Some IoT Applications



Good Example from L. Bononi

● One typical evening planning next working day...



Tomorrow first office meeting at 8.30am

Typical car trip in these days: 1 hour time

45 minutes to wake up and get ready...

I decide to set my alarm to wake up at 6:45am

What could (will) possibly go wrong?

- 1) At 4.30am it starts snowing
- 2) Truck obstruction along the usual path
- 3) Traffic congestion on alternative paths
- 4) No parking at destination
- 5) Bathroom cold when having shower
- 6) Coffee cold when having breakfast
- 7) Left my car keys at home when in garage
- 8) Elevator busy when leaving my flat
- 9) Total time to get ready+breakfast: 55 min.

Leaving 10 minutes late + 30 minutes additional travel time

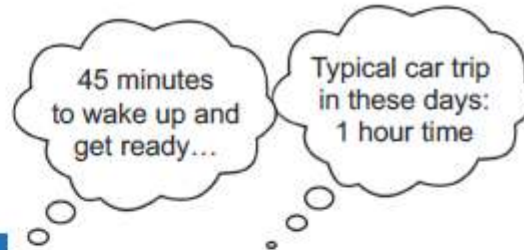
I missed the morning meeting!

Good Example from L. Bononi

● The IoT scenario...

This is the PROGRAMMING effort!
Define your user's objective!

Smart IoT Services



Based on my typical habits
decides to set my alarm
to wake up at 6:45am



What could (will) possibly go wrong?

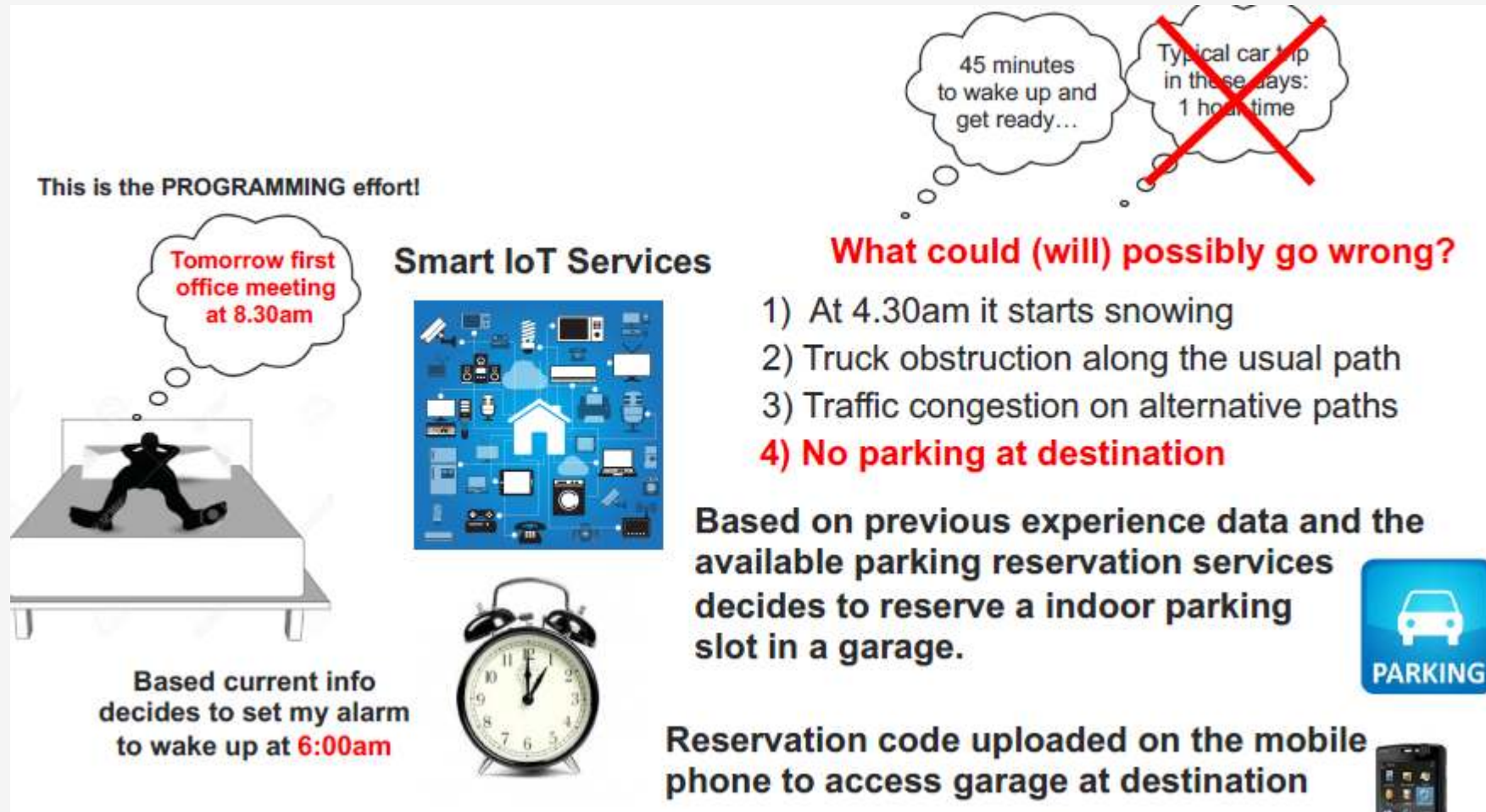
1) At 4.30am it starts snowing

Get notified in real time by the
weather monitoring system
or device.



Based on forecasts it anticipates the
alarm clock to 6:30am (15 minutes before)

Good Example from L. Bononi



Good Example from L. Bononi

● The IoT scenario...

This is the PROGRAMMING effort!
Define your user's objective!

Smart IoT Services



45 minutes
to wake up and
get ready...

ALARM activated at 6:00 am



Get message on smart phone
explaining the reason why the alarm
has been anticipated.



5) Bathroom cold when having shower

HOWEVER, 10 minutes before at 5:50am...
started warming up the bathroom to 23 degrees

6) Coffee cold when having breakfast

The mirror notifies I am leaving the bathroom, and
while I get dressed in my bedroom, the
COFFEE MACHINE is activated in the kitchen, and
the warming up of the bathroom is switched off.

When I leave my home passing through the main
door I got a message on my phone advising me
that I do not have the **CAR KEYS** with me...



When I re-enter the door my phone informs me
that the keys are in proximity of the lamp on the
table in the living room, and the **LAMP** flashes to
help me to find the **KEYS**

Good Example from L. Bononi

● The IoT scenario...

This is the PROGRAMMING effort!
Define your user's objective!

Smart IoT Services



45 minutes
to wake up and
get ready...

ALARM activated at 6:00 am



Get message on smart phone
explaining the reason why the alarm
has been anticipated.



I am leaving with my car right in (planned) time,
with my path already set in the navigation system
getting alerted of any problem on the path and
need to make detours in real time.

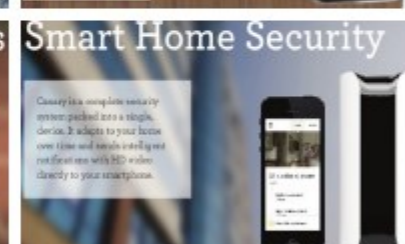
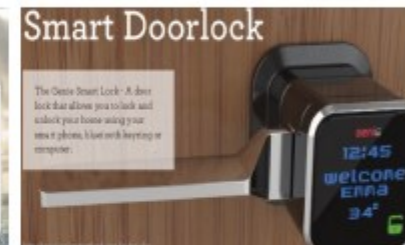


... and when at destination I will have my car
parked in reserved indoor garage with no delays
Barriers open with my contactless smartphone
code...



A galaxy of IoT objects already available today...

IDC: 50-220 Billions things connected in 2020-2025
IDC: market volume of 14.4 Trillions

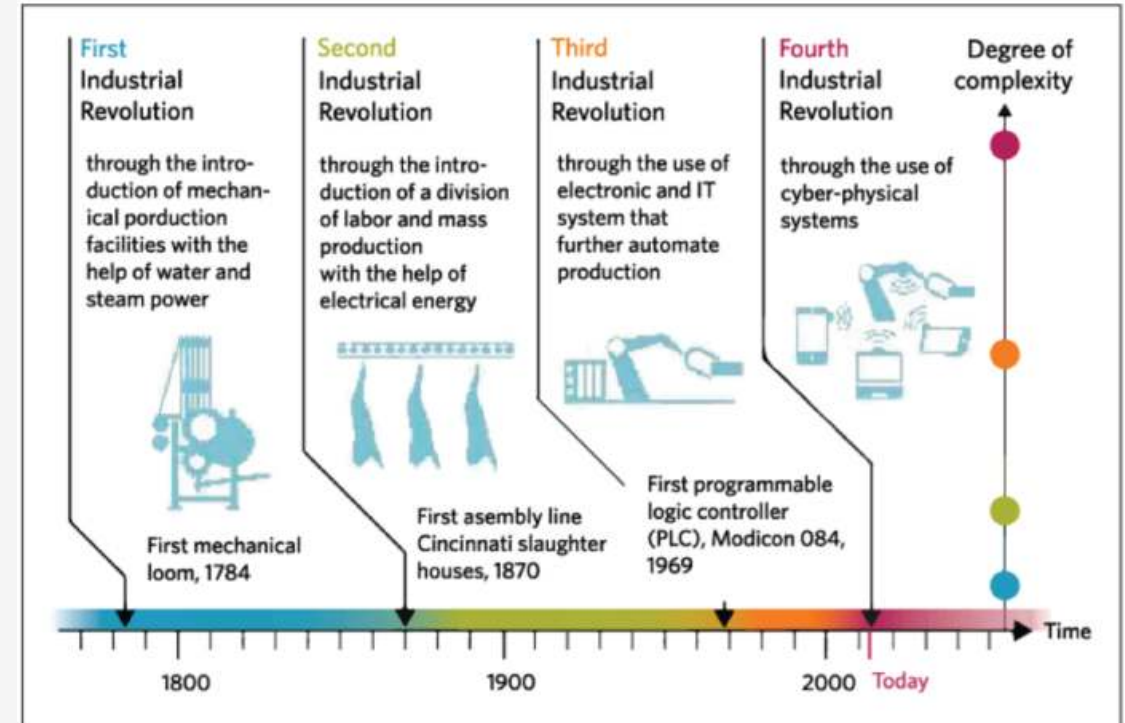


lot in Industry (Industry 4.0)

...so, why not to adopt these new paradigms in the industrial world?
Could we expect some benefits?

Industry 4.0 !

- to sense the environment and gather observation data
- to realize a new paradigm of data analysis and control
- to push automation, scalability, efficiency of processes
- to reduce costs/overheads and increase revenues and business
- etc.



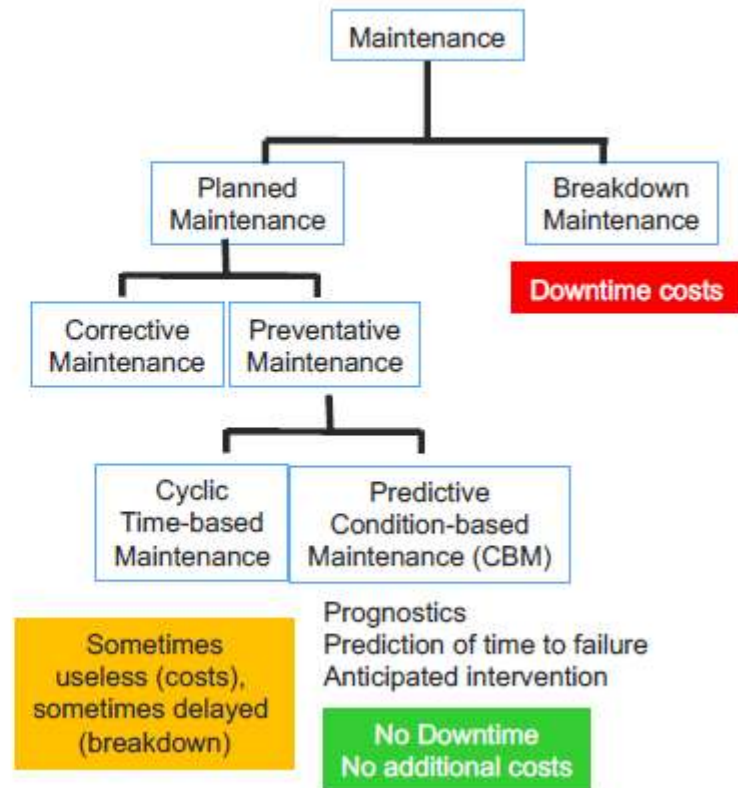
e.g. Predictive analytics and maintenance

Some concrete examples:

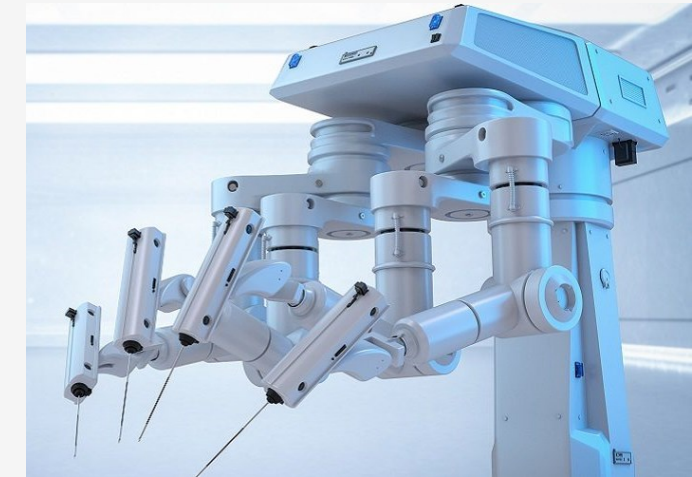
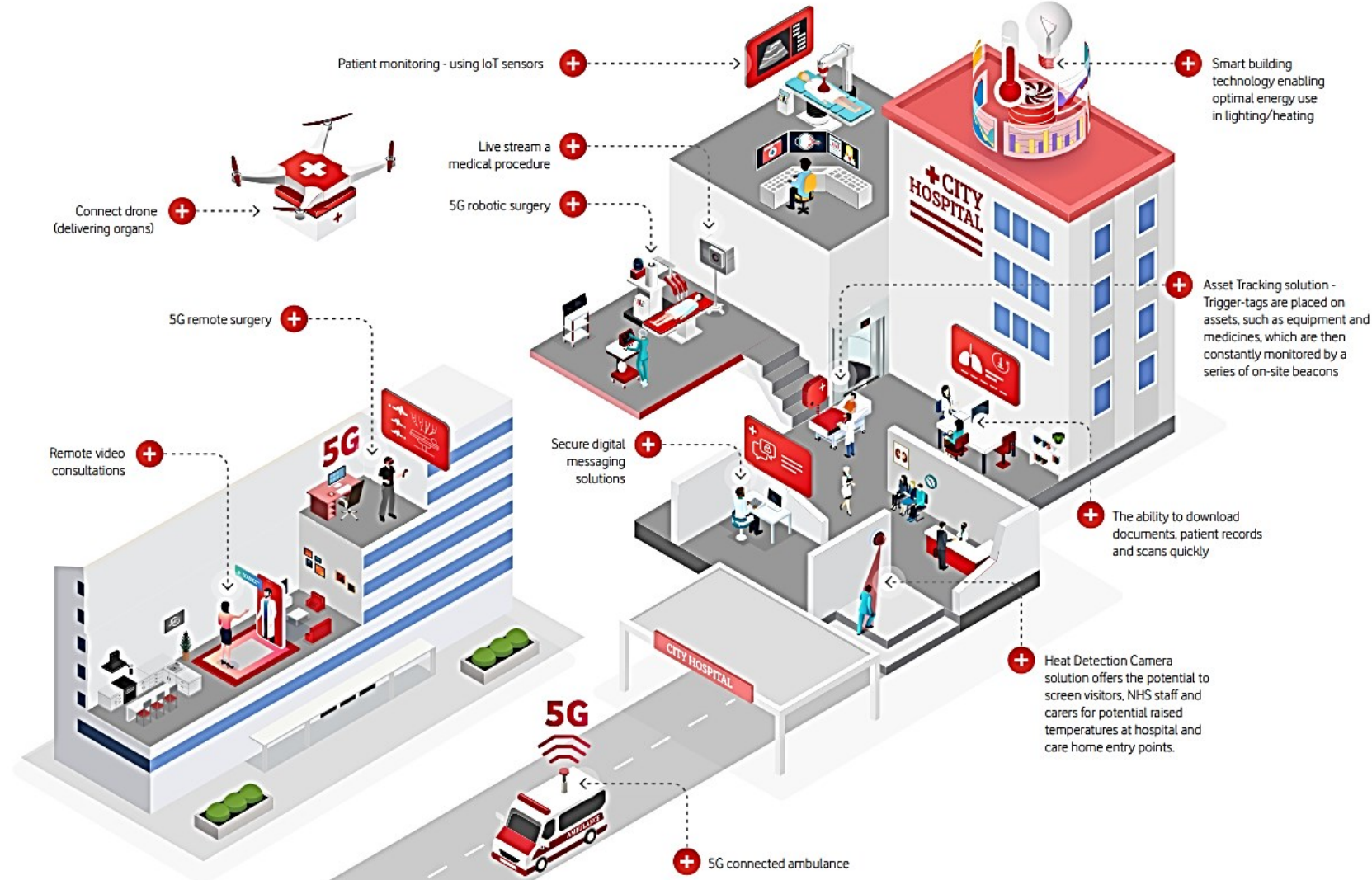
Rolls-Royce can now identify **correlations between different part failures** and different operational environments. This is allowing the company to **predict engine failures** several days before they occur, with high accuracy and low false alarms.

General Electric announced recently that revenues from its Predictivity solutions, which make machines **more productive**, exceeded \$1 billion in 2014.

Solair predictive maintenance as a fee-based **customer services** to AEG's household appliances (via app)

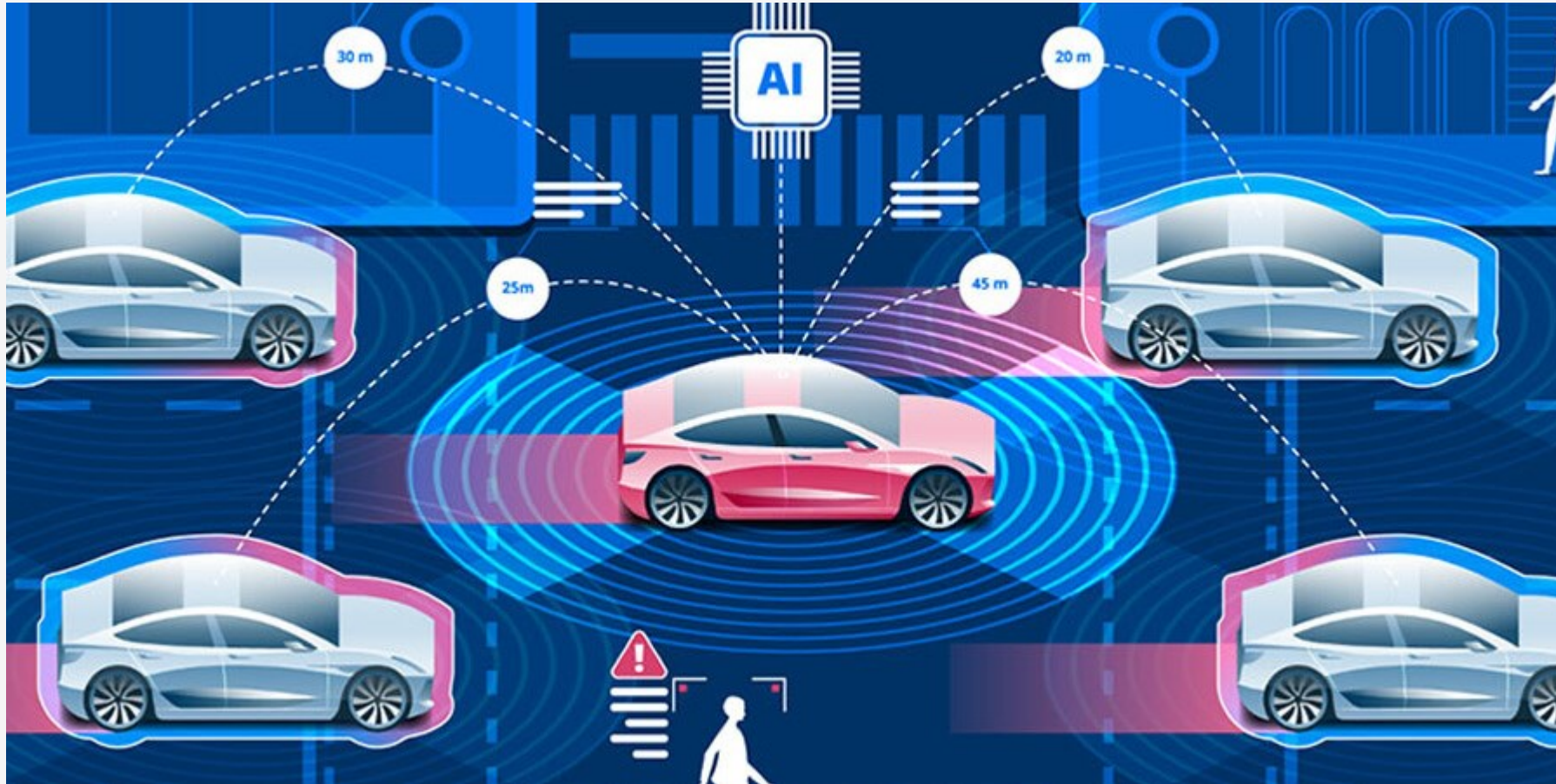


Some Applications: Hospitals in the future



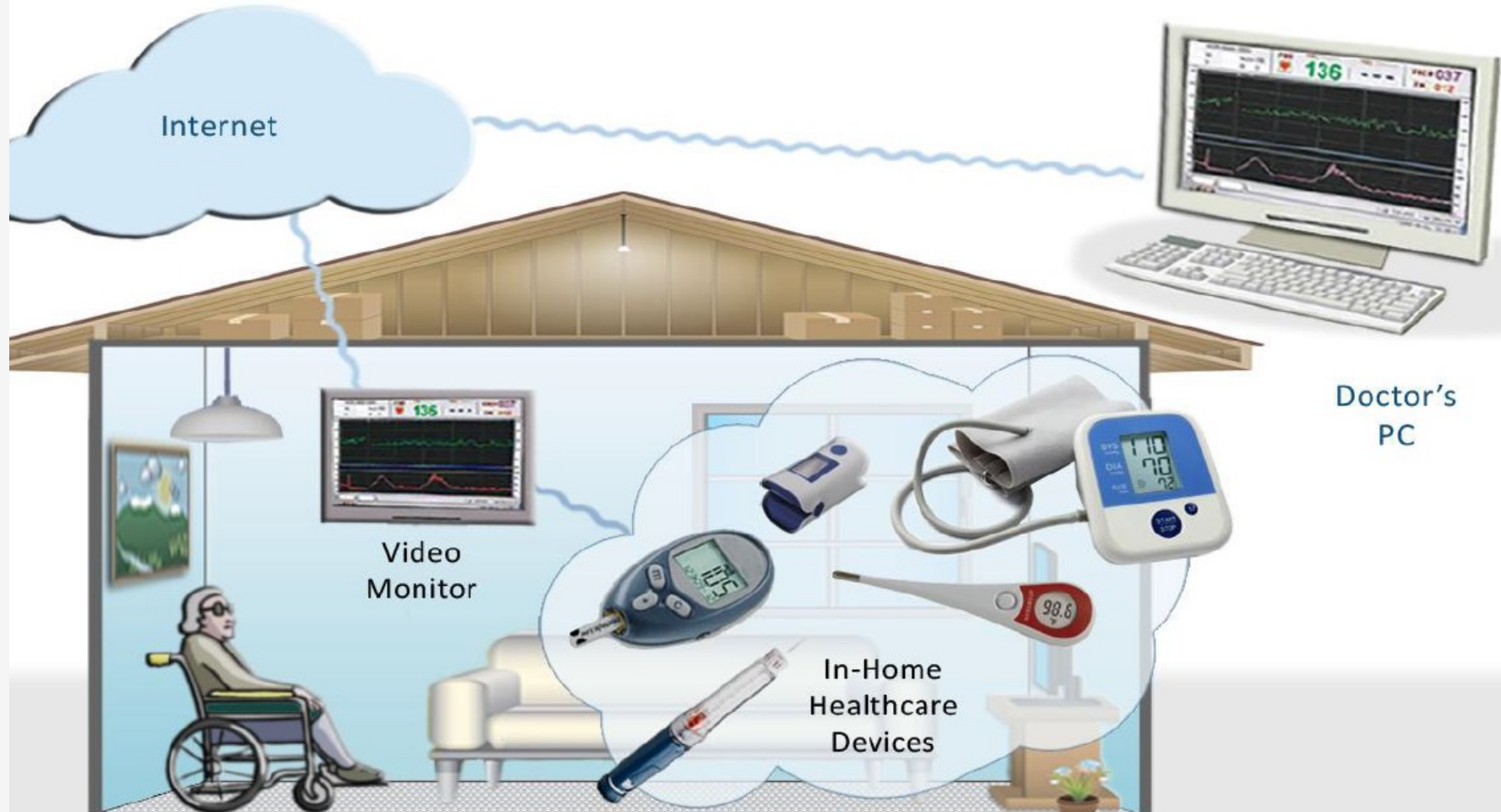
(Credit Vodafone)

Some Applications: Smart Vehicles & AI



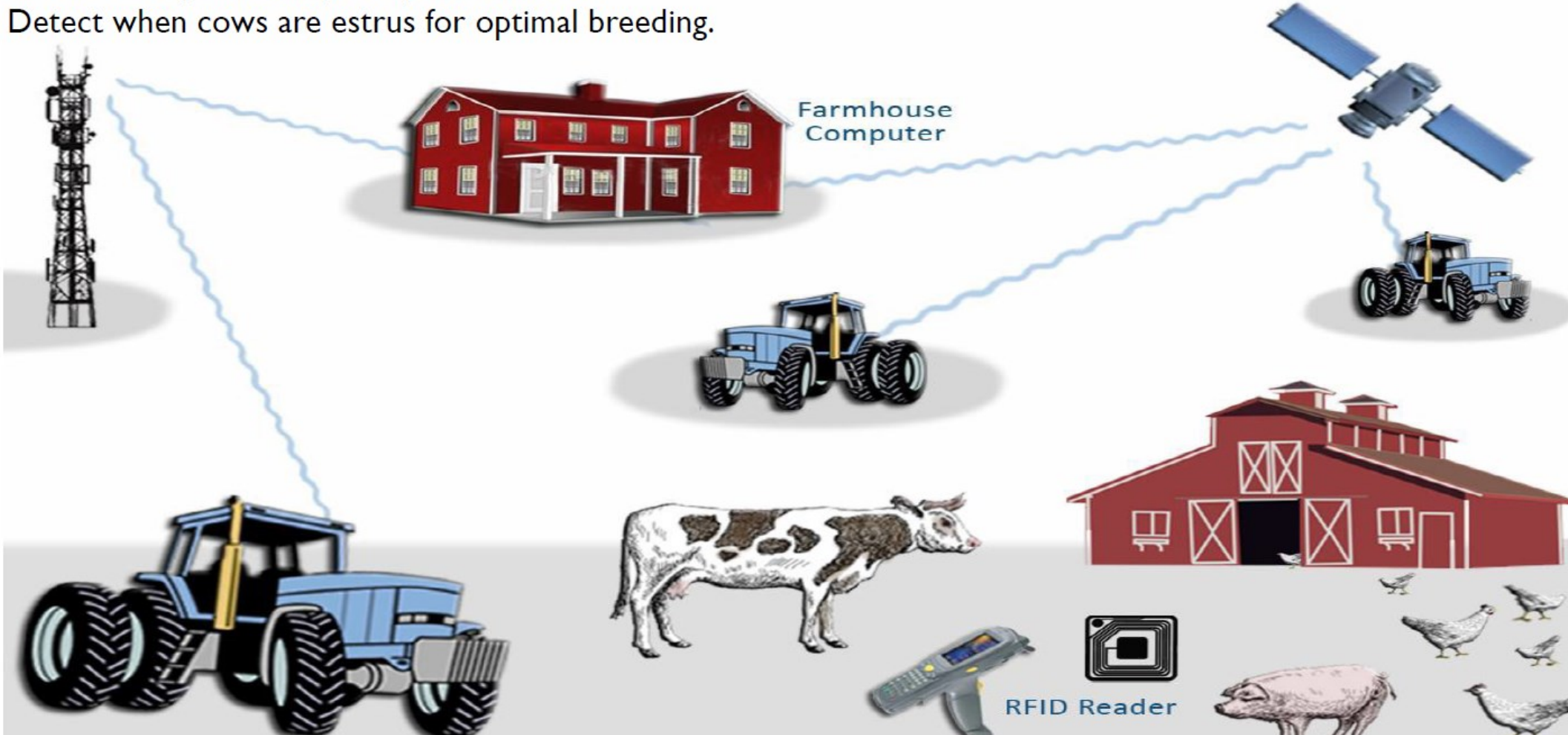
Some Applications: Health Care Systems

Offer remote health services for baby boomers.
Help them to live independently at their homes instead of nursing homes.

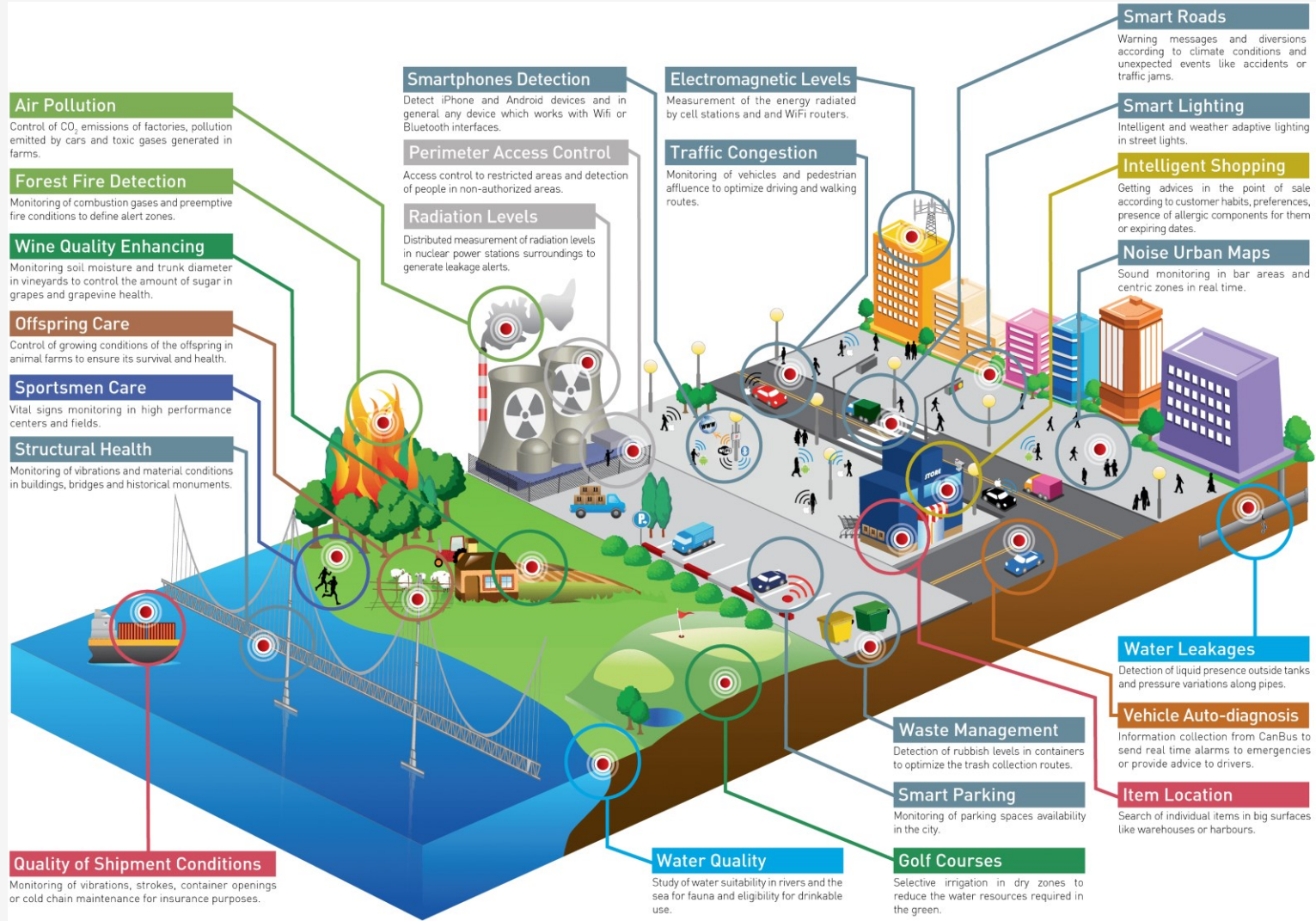


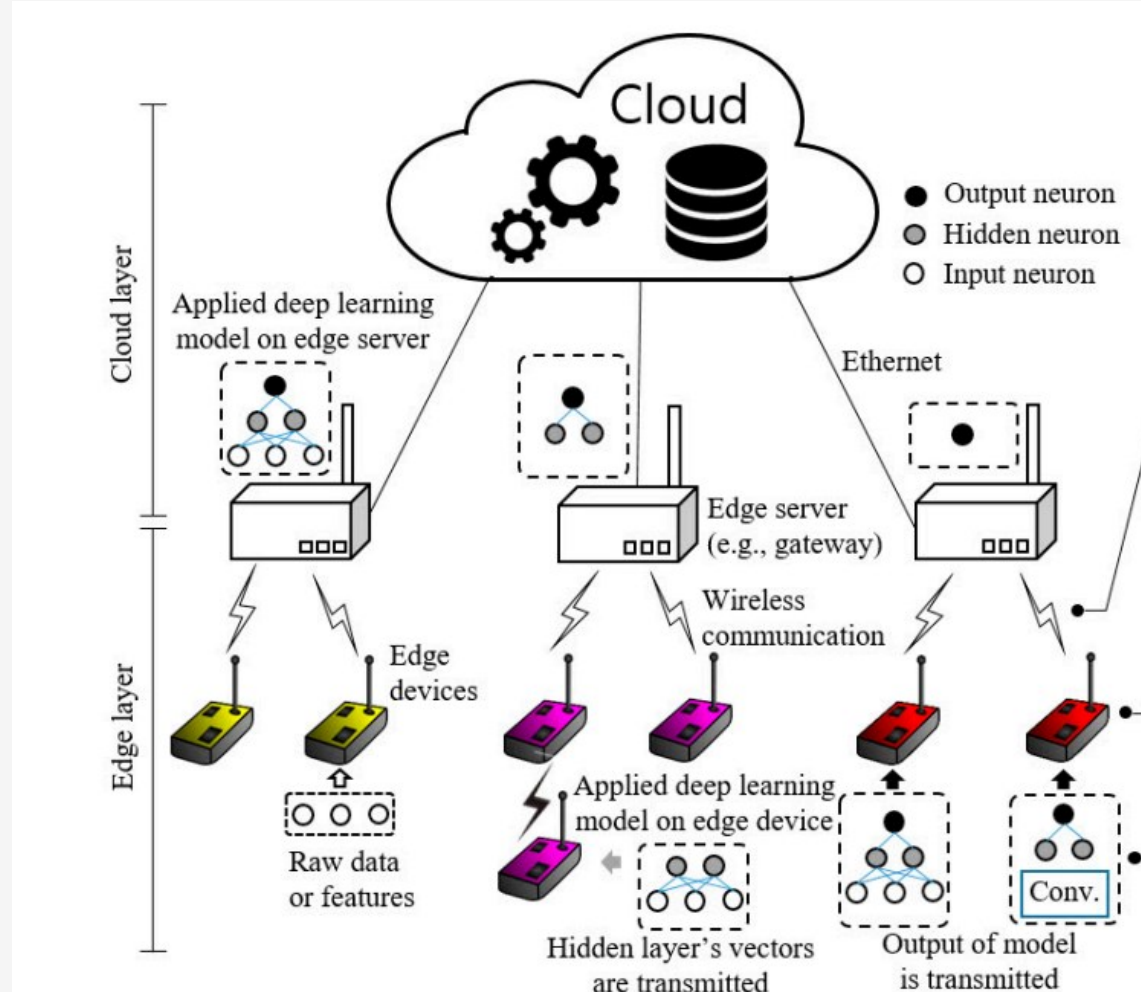
Some Applications: Smart Farms

- Temperature sensors, moisture sensors, etc.
- Sensors to trap and analyze captured insects.
- Detect when cows are estrus for optimal breeding.



Some Applications: Smart Cities





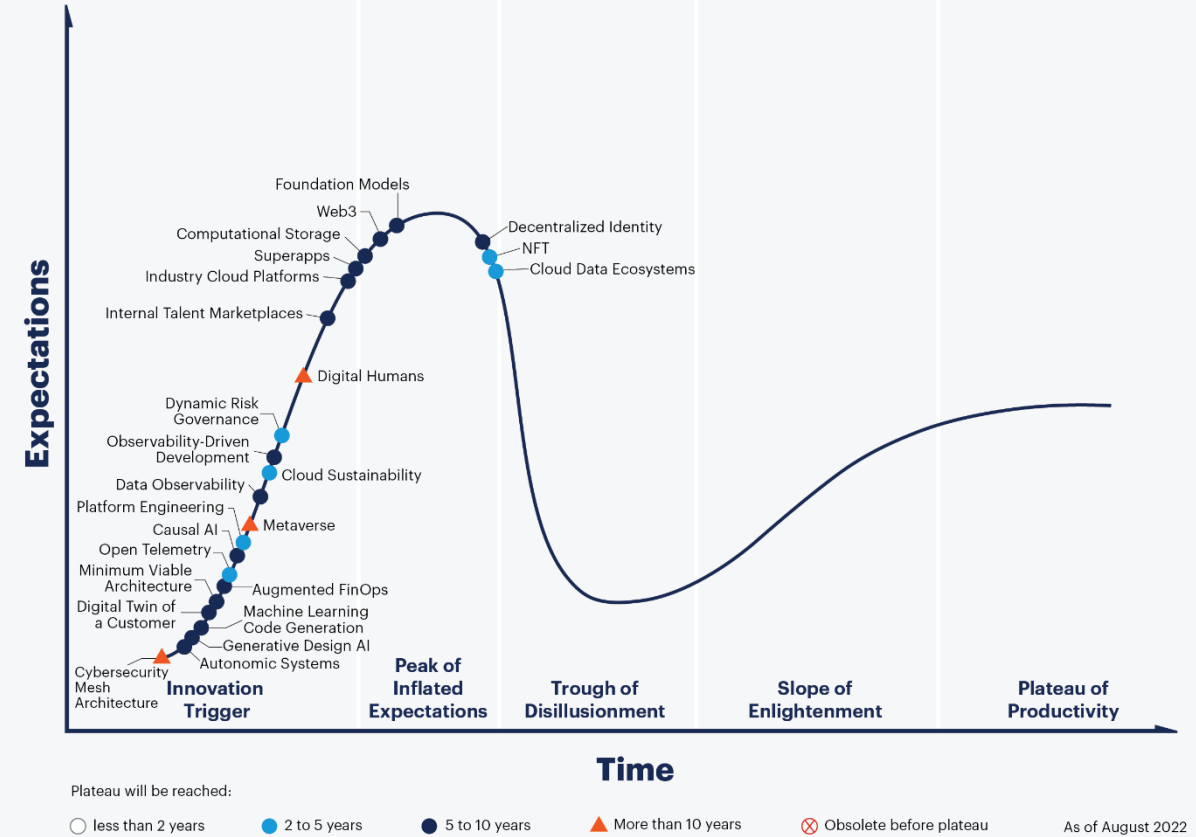
Kang, J., & Eom, D. S. (2019). Offloading and transmission strategies for IoT edge devices and networks. *Sensors*, 19(4), 835.

Last Slide for You

Yapay Zekâ Teknolojilerinde Öncelikli Ar-Ge ve Yenilik Konuları

Az Etiketli Veriyle Modelleme	Performansı Yüksek Pekileştirmeli Derin Öğrenme	İstatistiksel Üretken Modeller	Açıklanabilirlik Düzeyi En Az 4 Olan Derin Denetimli Öğrenme	Alan Bilgisine Sahip, Verimli, Güvenilir Modeller	Akıllı Robotik
Planlama ve Tahmin	Gerçek Zamanlı Görüntü İşleme	Doğal Dil İşleme (NLP)	Türkçe Konuşma Tanıma/Sentezleme	Federe Öğrenme Platformları	Diferansiyel Mahremiyet ile Verilerin Anonimleştirilmesi
Biliş ve Davranış Analizi	Yerli AutoML	DLaaS	Teshisi Zor Hastalıklar	Yardımcı Sağlıkta Açıklanabilir Yapay Zekâ	Potansiyel Hastalık Belirleme/Takip
İlaç Geliştirme	Bulaşıcı Hastalık Tespiti	Otonom Sürüler	Çok Amaçlı Otonom Robot	Kestirimci Bakım	E-Öğrenme ve Ölçme
Enerji Kayıp/Kaçak Ölçme Sistemleri	Finans, Ticaret ve Lojistikte Verimlilik Sistemleri	Tarım ve Hayvancılıkta Kalite ve Verim	Uydu ve Hava Görüntü Tanıma Sistemleri	(Mobilite) Otonomi	

Hype Cycle for Emerging Tech, 2022



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